

Provost's Lecture: Karl Friston on "I Am Therefore I Think"

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[Music] I'm really delighted to see that we have you know filled the entire house so welcome one and all welcome to the this iteration of the 2019 Provost's lecture series and today we're thrilled to welcome professor carl Friston the principal research fellow and scientific director of the Wellcome Trust Center for neural imaging as well as professor at the institute of neurology at University College London and a consultant at the National Hospital for neurology and neurosurgery in the United Kingdom so this talk is sponsored not only by the professor or the Provost's office but it's also sponsored by the Center for neural circuit dynamics at Stony Brook University the Department of Neurobiology and behavior as well as the department of psychiatry at Stony Brook University so welcome Carl on behalf of all of us and we're delighted to have you here so the and as well we are welcomed by my own organization the Institute for Advanced computational science of which I'm the student president and the ICS the Institute for this you know advanced computational science really tries hard to make sure that we invite speakers whose work touches on a wide variety of perspectives and Carl really embodies that that ethos so it's really hard to define an area of research where Carl's methods have influenced right so Carl for example is the inventor of statistical parametric mapping voxel-based morphometry and dynamic causal modeling okay all motivated by schizophrenia research based mathematical contributions include variational laplacian procedures and generalized filtering for hierarchical Bayesian modeling version as well as contributions in theoretical neurobiology his most famous contribution being the free-energy principle for action and perception which I expect will feature heavily in today's talk okay professor Friston also holds honorary doctorates from the universities of York Zurich and Radboud University he's a member of the Academia Europea a member of the European molecular biology organization the Royal Society the Academy of Medical Sciences and he was president of the International Organization of human brain mapping he's the recipient the

first recipient of the young investigators award in human brain mapping he's a recipient of the Minerva golden brain award we see the metal from the collège de France received the Wheldon memorial prize and medal in 2013 for contributions to mathematical biology and it was the most recently he was the 2016 recipient of the Charles branch award for unparalleled breakthroughs in brain research as well as the glass brain award which is a Lifetime Achievement Award in the field of human brain mapping more recently Tristan's work has been on the models of functional integration in the human brain in the principles that underlie neuronal interactions so today we're very happy to welcome here to talk about this subject and more broadly on the topic of I am therefore I think so please join me in welcoming professor well thank you for that lovely introduction I told him he could say three things about me and he thirty things always a pleasure to hear them though thank you there's also a great pleasure to be here you you have one of those beautiful parts of America I've ever visited you must be gorgeous working here I've also enjoyed meeting hundreds of you I have to say I'm absolutely worn out now I've met so many interesting people there are so many interesting things have been asked so many difficult questions that forgive me if my brain starting to slow down for this lecture this lecture is really a survey a back story to a trend in the cognitive neurosciences over the past decade or so which is a move towards understanding the brain as a statistic organ a predictive processing machine so what I want to do is just cast for you the main elements of the back story as would be articulated in the first half of a talk by physicist and in the second half of the talk by a neuroscientist and I'm doing that because I am aware that I'm speaking to a very eclectic and possibly informed more informed than usual audience so I will have equations in case you're worried it's going to be two back story issue so here's the the overview of the talk I'm going to discuss the properties that systems must possess if they exist and by exists I mean show some form of X distant existential invariance of the soul we associate with living systems I'm going to illustrate and demystify some of those principles using numerical analysis simulations of a primordial soup just to evolve a little creature and ask some questions about the nature of its behavior and how it and then return to the point of view of a neuroscientist and I'm going to tell exactly the same story but this time as a neurobiologist and talking about the anatomy of inference bringing predictive coding implementations of these principles and how they might be implemented by neural networks and then finally if we have time and I got to get a helpful hurry up sign from my host if we run out of time I'm going to justify the use of the quirky title I I am therefore I think I think therefore write out whichever you prefer so we're going to make the final move which is really

rehearse in the same principles but now under a prior belief that I am the sort of creature that exists and therefore must conform to these principles and we'll conclude with simulations of visual foraging and epistemic affordance so let's start I'm going to start with a question posed by Schrodinger how can the events in space and time which take place within the spatial boundary of a living organism be accounted for by physics and chemistry I'm not going to answer that question clearly but we'll pick up on the notion of a spatial boundary and just note that in order to talk about anything you need to be able to demarcate or distinguish disambiguate the thing from something else or nothing and that really is the role of the boundary that separates the thing from the non thing and scrolling would be the first person to acknowledge that this boundary was in and of itself a probabilistic stochastic quantity and so I'm going to take that boundary to be a Markov blanket so at this point I normally ask to put your hand up if you know what a Markov blanket is good can you tell your friends what a Markov blanket is now you could have been clever a news excuse you haven't got the microphone for the recording but I think it's just shyness and what he would have told you what he would have said was that well let's take this graphic here let's assume that this is a description of some little universe and all these circles here denote states of any abstract sort and the edges the arrows denote an influence of one state over another state and let's just take some internal states of interest and the Markov blanket comprises those other states that in a statistical sense insulate blanket render insula the internal states from the external states and by that I mean that if I wanted to predict the internal state based upon the rest of the universe I would only ever need to know these blanket States and technically they're defined in terms of the parents the children and the parents of the children of the internal states of interest so a very simple definition just in terms of conditional dependencies or influences or some adjacency matrix on this graph that provides a set of states that separates internal from external now only about 25 percent of you put your hand up and I said what is a Markov blanket if I you do all know what a Markov blanket is so any of you that have worked with time series or hidden Markov models and Markovian processes have implicitly been using the concept and know the concept of a Markov blanket so if you just think of these states being states of the universe in time then the Markov blanket is just the present moment in the sense that everything I need to know about the past that informs the future under a Markovian assumption is in the present so the present provides a Markov blanket that statistically separates the past from the future so that's all this means all that we're doing here is generalizing that notion to any our basis arbitrary set of states that can be articulated in two of

their dependencies and influences in this forum here I'm going to make another distinction which will become important later on and I'm going to make a bi partition of this set of states here and into sensory states that are not influenced by internal states and active states that are not influenced by external states and just to motivate that distinction what I'm going to invite you to do is just to think about these conditional dependencies this causal statistical architecture in the context of things that we all know and love and sometimes studied so these blanket states comprise the active and sensory states that separate external from internal states and what I've done here is just rewrite that graphic that dependency graph in a form that you that as a neuroscientist you recognize as a cellular biologist you recognize where we can associate the internal state and every state of the brain there I'd need to list in order to say this is a state in the present moment all the synaptic efficacy and activities and connection strengths the active States could be the states of all my muscles my autonomic reflexes my motor system that cause changes in the external states that produce sensory impressions upon my sensory states my cell 3 epithelium that subsequently change my internal States to induce changes in action and so the cycle unfolds and I use the word cycle deliberately here because what we have is just a statistical statement of some form of action perception or perception action cycle exactly the same dependency structure can be found at the cellular level so we have all the intra cellular states that could constitute a little cell here we can associate say the active space with the actin filaments that support and provide motility to the surface or sensory states of a cell with receptors that are in receipt of influences from the external meliha that in turn affect the internal states that affect the cell motility and this structure just rests upon the absence of two conditional dependencies that define the active and the internal state an absence of an influence of internal states on the sensory states and an absence of the influence of the external states on the active States and that's it that's all you need to define something that can be separated from something else so what and then to do now is to ask you just to forget about the Markov blanket for a moment what we're going to do is just quickly run through the dynamics of stuff that exists and then what we're going to do is to put the Markov blanket back in play and see what interpretations follow from the existence of a Markov blanket under more generic principles of self-organization so one of cartoon here is a state space there's two states here and we have some system say me and I mean one state at any one particular time and I trace out a trajectory or a path through my state of my face space here and as time unfolds I habitually revisit or systematically revisit neighborhoods of states that I have been in before so you can think of this say for example I'm getting up in the morning I'm

having my cup of coffee and do my emails I do the morning work I have lunch and so on this could be my daily cycle or it could be my cardiac cycle with systole and diastole as I circulate around this attracting sentence attracting manifold crucially returning to the same state of being periodically in a very itinerant way this enormously complicated very high-dimensional attracting set but it must exist in the sense that I exist in the sense that I have measurable states that I keep on returning to define the sort of system or the thing that I have technically this would be a pull back or around and dynamic attractor I'm just going to refer to it as a set of attracting of states that must be there if something exists now the reason I'm going on about that is that there's a lot of machinery which we serve as we'll know and love that is apt for describing some of the fundamental dynamic so that must be a play if this attracting set is in plane so here I've just written down the long-run description of this system in terms of motion which is equal to flow dependence upon States but some random fluctuations Omega I can describe not just the flow on this attracting set but interpret this flow in terms of the probability that I would be in this state if I was sampled at random at a random time in terms of a probability density and I'm going to associated that probability density with the the sorts of states that I'm in when I have attained or settled down to my crafting set when I am in non-equilibrium steady state and if I do that that means that this probability the probability of being in a particular state given me is described by the density dynamics ready to change the probability of my States given me and I can write that down using the fokker-planck equation the master equation the congrats forward equation whatever you want to describe this interestingly it will transpire also to be this floating away the equation and I can write it down as a function of the amplitude of the random fluctuations and the gradients of this and the flow and the probability distribution if you don't know what this means don't worry about it it's just something that has to apply if this attracting set or not non equilibrium steady state exists the key thing about it is that I know but once I settle down this is zero and if this is zero that now allows me to write down the flow as a function of the probability the gradients of the log probabilities of states of being that I am typically in given the amplitude of the random fluctuations and a solenoid I component here Q which is anti-symmetric matrix so this is gonna be quite crucial it's basically as if I exist my dynamics my flow must satisfy this equation and this equation is a functional or a function of a log probability we is just the sorts of states that characterized me for those of you who don't know this formalism I'll just very quickly go through in a mock in a more intuitive fashion imagine that I place a drop of ink in a cup of water and due to the random thermal fluctuations the molecules of that ink will disperse

them cells throughout the solvent so I would expect to see for a system that wasn't going to attain any form of steady state equilibrium or otherwise the ink diffuse until the concentration is uniform everywhere in the solvent that is not the characteristic of the systems that are interesting as what we are talking about the systems that do this I drop the water in it and then it seems to the oil ink and it seems to gather itself up so attains its non equilibrium self-assembly as if the molecules are flowing up probability gradients so I can reverse diffusion and it is exactly that particular dynamics that that solution to the fokker-planck equation describes so if we associate the concentration of the ink molecules with the probability distribution then what we're saying is we can understand these two components of the flow as basically this gradient flow climbing up probability or concentration gradients and this solenoid 'el or divergence free flow as flowing around the ISO contours here and it's in particular this gradient flow that I want to drill down it is there in order to exactly balance the dispersion of the random fluctuations so I'm not talking about detailed balance I'm talking about something much more general that if I have attained non equilibrium steady state and I exist in that sense then it must be that on average my flow is exactly countering the dispersion of my non-equilibrium statistical T distribution so that's good is it interesting it is very interesting as a lightly the godfather of most of physics so that solution and in particular the fokker-planck equation under right so you can coerce it to underwrite classical mechanics and quantum mechanics so for example we can set the amplitude of those random fluctuations to zero and what falls out of that solenoidal flow because the gamma part of the gradient flow has gone away is just classical Lagrangian mechanics and Newtonian laws we can go to the other end of the scale and ignore the solenoidal bits and just focus on the random fluctuation and all the interesting behaviors that one gets from random fluctuations that do not cannot see these probability gradients they just pass through potential barriers for example and then what we get from that is basically a a formulation that ends up with things like the fluctuation theorems modulation dissipation theorem and stochastic or for ensemble statistical mechanics we can just look at the square root complex square root of that non equilibrium state density and treat that as a wave function and then the fokker-planck equation writes itself as a as a spur dinner equation what we're going to do though is consider another sort of mechanics that resps explicitly on the Markov blanket so that's the move here we're just taking standard physics and saying what would it look like if there was always a Markov blanket there and that brings some interesting things to the table first of all it means that there are two sets of dynamics dynamics that pertain to the internal states and the active States that look as if

they're doing gradient flows and they have a purpose and furthermore we're going to interpret a bound on the thing that they are flowing on in terms of a variation free energy so that's the basic story I'm just foreshadowing or for grounding that story as an extension the entirely consistent with and complementary to things that we are be taught at school so that's our gradient flow the solution to the fokker-planck equation and this is the same solution but now just written down in terms of the states of a particle we're assuming that a particle comprises the internal States and it's Markov blanket that has active and sensory states and I'm focusing on the active States here and because of those Independence's I only only need to concentrate on the probability of the blanket States the sensory inactive states so this is a solution to this non-equilibrium steady state there must be they're describing the flow of internal and active States and the interesting thing here is the both look as if they're trying to optimize or maximize the same quantity and that's the log the gradient of the log probability of the blanket States given the system me in question and the story is going to be basically that we can look at the this mechanics as perception and the flow of active States in terms of action and what licenses me to say that well just the form of that equation makes a lot of sense from the point of view of many many different perspectives in psychology and the life sciences so I'll just briefly run through a few of them just and you will know this stuff for at least one or two of these things this just so you can see how I've joined the dots from the point of view of motivating the central importance of this long probability up here so we've just said that these describe attracting States so the greater the probability of being in any attracting state or blanket state here means those are the sorts of states that I've attracted to that I will be typically found in basically they can be construed as having value for me the things that I aspire to or they're attract me and what we're saying is basically that both perception and action are in the game of maximizing value and on that reading we can spin out things like reinforcement learning control theory until if we're doing economics it could even be expected utility theory that's nice because the negative of this value is known as self information and information theory tribus call it surprisal I'm just gonna refer to it as a surprise it's as the empower bility of a negative log probability of being in some state incidentally that's the quantity that's going to be upper bounded by the free energy so I'm going to use free energy synonymously really with this surprisal or surprise so what this says is the systems that possess a Markov blanket and have attained non equilibrium steady state are basically trying to get and put an upper bound on surprise and from that we can understand things like the informatics principle principles a minimum redundancy and indeed the variational free energy

principle that we're going to be talking about that's nice because this thing here when taken in expectation or the average of this over time or over the path is entropy so what we're saying is that this dual minimization of this quantity here this surprisal is just a statement of self-organization it's putting an upper bound on the dispersion or the entropy of the disorder of me and that's how you might understand it as a physicist or from the point of view of synergetics because if you're a physiologist it's just a statement of homeostasis that's all we're saying we're just saying that systems are persevering in a changing world are those that keep their physiological states within viable bounds they gather themselves up and place bounds upon that dispersion the final interpretation which is the one that I want to leave region is from the point of view of a statistician this probability here is it's the probability of some sensory or blanket data condition now not upon me but if I treat me as a model of those data this now becomes something called model evidence or marginal likelihood and it is the holy grail in terms of the quantity that you want to optimize when doing Bayesian inference and if we take that interpretation then we what we are saying is that this is equivalent to say that we can understand said the brain did this is a Bayesian brain we'd understand that in terms of evidence accumulation and indeed predictive coding and that's the sort of story that I'm going to try and to tell you for the remainder of the talk before I do so that is I promise you that I try and demystify some of the behavior entailed by that formulation a non-equilibrium steady-state using simulations so here the simulations what I've done here is take a hundred and so if a macromolecule synthetic macromolecules equipped them with autonomous dynamics I like a little creating a little acted super acted matter and then coupled them with strong and weak forces strong repulsive forces and depend upon the spatial distance and weak electrochemical coupling here and just let them self-organized and fall settle down onto their attracting set and once I've done that I've created a little universe where crucially I know all those dependencies and I know all of the dependencies that would otherwise define on that would define the Markov blanket so I can now go in there and search for the Markov blanket and ask is there a little creature living in the middle of this little soup and if I identified this the eight most connected states I can then using the dependences incredible adjacency matrix a find that parents the children and the parents and the children and identify this little synthetic creature with its internal states encoded in blue the active states surrounding it encoded in red that support the sensory states of magenta that then separate the internal states and external state since I on and it sits there happily whittling away wagging its tail could be like a little there cilium here now I've simulated that thing that has a Markov blanket I can now ask well is that

self that Bayesian interpretation of the dynamics that must be there given is now settled down to non equilibrium steady state is that a viable description of the behavior of this little synthetic creature so in other words are there any attributes or states of the configurations of internal states that in some sense model or predict the states on the outside and it's very simple to show that yes indeed that has to be the case as is the case so here's our a numerical example of that what I've done here is just taken a mixture of time lagged internal electrochemical States and ask what mixture best predicts some mixture or some motion physical motion of the external states and these are the internal states and here in the solid line is a prediction and the dotted line is the actual velocity of these external states so if you're a you're a scientist you can imagine this is a simulated visual motion experiment you know are there any internal electrochemical fluctuations that predict or are predicted by changes in visual motion out there that I cannot access directly but I can do through my sensory epithelium my retina and you can see that not only is our correlation but you can actually pick up events and even see them in terms of the internal fluctuations here is highlighted by the white arrow the reason I've shown this example is if you look closely you can see that the internal States actually predate this rapid expulsion of this external state a little bit like a solar flare and then it's pulled back again so the internal states are not really seeing the external states it looks as if they may also cause in terms of temporal precedents the external states that may be causing the internal state so the question is what's causing what let me ask a quest let me pose others question you know the internal space causing the external states or the external states causing the internal states what it is very much a tricular leg we do which side do you want to come down the very fat you're teasing me with a flippant answer means you know the right answer which of course neither it's good this is this is just an instance of what Huygens noted centuries ago of generalised synchronization or chaos our generalized synchrony so this is the story where you have old-fashioned clocks that you are suspended from the same wall or the same beam and inevitably they're attracting sej requires them to be on the same synchronization money fell so they will eventually come to swing in synchrony inevitably and that's all that we're seeing here we're just seeing a completely symmetrical synchronization of dynamics in virtue of them both the outside and the inside being conditioned upon the Markov blanket it can be no other way which is an interesting picture so in fact this is a picture that he actually drew of his two clocks with the beam here so from our point of view one clock would be the internal states these cover the external space and the beam is the constituting the blanket States and one interesting aspect of this is it shows you the symmetry of the mathematical argument here so we I have

had an interesting conversation this afternoon what this symmetry means and what it means is if you subscribe to the view that the dynamics here are all about inferring what's going on over here then the converse has to be true that means that the environment is trying to infer and to learn about you as much as you are trying to learn about the environment and our conversation went right through to evolution and eco niche construction so it's not a silly view and it's certainly a mathematical truism anyway so that's all the heavy lifting that's all the hard part done just to summarize what we said so far the existence of a Markov blanket necessarily implies a partition of states into internal states their blanket sensory inactive states an external or hidden states hidden behind the blanket and because active states change but are not changed by external states they minimize the entropy of the internal states and their blanket and this means an action will appear to maintain the structural and functional integrity of the blanket and some people have written about that in terms of auto police's or self-assembly unlike internal states appear to infer the hidden causes of sensory states by maximizing Bayesian model evidence and influence those cause causes through action and I'm going to refer to that as active inference so just different takes or losses on something that is necessarily there if you exist and you have a Markov blanket no I've forgotten about this is something that one of my visiting research fellow said me not last weekend the weekend before so for those of you who think the Markov blanket is some abstract mathematical constant it is not you can buy them in America so this is this is Frances little baby Kira to two months since two weeks old now and this is what his wife bought him so that's actually Markov circuit 1858 I think keeping her states warm as she makes her own inferences okay so as promised what I'm going to do now is just rehearse but using a different rhetoric this same story as if I were giving a lecture or talking to neuroscientists and I'm going to now say well what this means or what it would look like is the things that have brains will appear to be little constructive organs statistical organs that will all be always appear to trying to infer predict model what's going on beyond their sensorium and this is beautifully illustrated by this it's yeah 16th century oil painter who was known for painting still lines that when viewed from another perspective have a very different interpretation you immediately have a very different hypothesis about what caused that particular pattern of sensory input so if you now see a face there where you didn't before the key thing is you made that that face was on the inside so the argument here in contrast sort of 20th century view of perceptual synthesis and in particular some visual inference is it it is not an outside-in process it is not as a series of clever filters or extraction devices it's very much an inside-out process that you have a hypothesis that's in there and you

submit that hypothesis to the sensory evidence and then reject or accept that hypothesis if you can predict what's going on that's the basic idea it's a sort of strange inversion of a Denon like sort of 20th century thinking about perception so it's a much more constructivist one that speaks to this little circular causality that was seen from the physical perspective so this idea is not new you can trace it back to the students of Plato so here are some of my favorite intellectual architects of this style of thinking probably best articulated by Helmholtz so for example objects are always imagined as being present in the field of vision as would have to be there in order to produce the same impression on the nervous mechanism so again what he's saying is you have to have the hypothesis the model the internal dynamics that would be able to predict what I would see if that was actually out otherwise you can't perceive it and this of course is very closely related to Richard Gregory's notion of perception as I pollicis testing then literally we palpate the world and we'll see an example this in the final slide with our eyes we visually palpate in the world to garner gather evidence that confirms or disconfirms our beliefs about what's out there causing our sensory impressions and these ideas have been taken the use of great effect by people in machine learning Geoffrey Hinton and Peter Dineen came up with the Helmholtz machine and borrowed the notion of variational Bayes operation free energy minimization developed by Richard Feynman quantum mechanics in the setting of a sort of Bayesian formulation of probability theory do you use these ideas that are still being used in various incarnations for example variational autoencoders in deep learning to great effect in terms of inferring on classifying and recognizing let's go back to Helmholtz and the notion of the the notion of an impression on the nervous system so for us that's this basically this sort of sensory impression on our sensory blanket or a sensory veil so if this infants perspective is right what this means is that we have to explain how our brains can take these sensory impressions and from them derive an explanation as to what course and so what's behind what's hidden behind this sensory blanket and of course the Amazon we've already got the answer it's just the solution to the complaint equation so what would that look like if you were a neuroscientist well I've actually written it down in a way that you'd recognize if you're an engineer doing time series analysis here so what I've done is written down this gradient flow that we've established as the solution to the the non-equilibrium steady state I've just swapped out the log probability with this free energy bound on the low probability that we don't need to worry about that at the so now we're doing a gradient flow on the variation free energy and if it's rewritten these two gradient and solenoidal components in terms of a divergence free update where μ now stands in for the internal states and are rewritten the

free energy here in terms of prediction error weighted by their precision on the squared prediction error weighted by their inverse variance or prediction here now the result done that is that that is a Kalman filter where the precision now plays the role of a Kalman gain and the way that these things work is that they gather evidence to infer the hidden states of the causes of sense data based upon a prediction given what they believe there so their expectations so let's take view now the internal states to encode a belief namely the expectation about what will happen next so this is a derivative operator so now describing the change in the states and an update term based upon a prediction error now what is a prediction error well let's just say we have this sensory impression here this shadow and we have this hypothesis encoded by our internal expectation say neuronal activity that it was caused by a howling dog and if I had a model or a journey of mapping between my expectation of the causes and the sensory consequences of that cause say gee to generate the consequences from the causes I could simply compare my predictions with what I'm actually sensing to generate a prediction error and that means that we can understand this gradient flow that must be there in terms of updating my expectations to minimize prediction error and if I can do that successfully I always then have an adequate explanation for my sensorial crucially it doesn't actually have to be the correct explanation so it might not be a dog but I will never know that and I never need to know that provided I could always keep my prediction error so my explanation for the world is perfectly or sufficiently good provided I can always get the prediction error precision where two prediction error minimize or upper bounded or minimize the generalization that which is this variational free-energy that itself is abound upon this surprising or self information so that's the basic idea it's nice because it provides its simple perspective on the intimate relationship between action and perception if everything is in the game of minimizing prediction error then there are we have a nice way of understanding the two complimentary ways in which we can do that first of all we can change our predictions to make them more like the sensory impressions and we can think of that as belief updating in the brain to improve our predictions to minimize the prediction error and any perception by changing predictions but there's another way of doing it we can make the sensations that we sample more like our predictions so we can actually act upon the world to go and literally palpate it in a way that selectively samples those sensory outcomes that we predicted a priori and it turns out that that looks exactly like classical reflex arcs both in motor control theory and also in other in terms of autonomic reflexes basically a servo that is provided with a neat like a thermostat just provided with a prediction it's setpoint and then all the peripheral machinery the action is just

in the service of changing the world bring the temperature back down to the setpoint just minimizing that precision weighted prediction error and that's basically we're going to try and illustrate in the final few slides and first of all just to say if this is going to work then we have to make realistic predictions of our sensory impressions and clearly the sorts of impressions that we are subject to are deeply structured very dynamic which means that our garroted models that are used to generate these predictions to form the prediction error are going to have to be in turn deeply structured basically deep models but also dynamic they're going to be deep dynamic your models and clearly that's a nice to hear if you're a neurobiologist because that's exactly how the brain is organized certainly it hits you in the face if you look at the visual system so I just khartoum that here so we have these two hypotheses or causes if we wanted to predict some retinal sensory data here you know what am I looking at and where am I palpating it and I put these two causes together and I'd cascade them down ultimately to make some predictions about the sensory flow that would actually be sampled if these explanations were correct so here the generative process these causes cascading down here to produce sensory states at the center of the Markov blanket plus the random fluctuations at each point so what would an inference machine what would self-organization non-equilibrium State look like based upon this architecture well it would just have the form of our gradient flow and when I write that down and swap out the way in which these sensory states were generated with the prediction errors and the predictions or expectations here what we get is a message passing scheme that looks very much like what we see in the rail brain so the random fluctuations have now been replaced by prediction errors and the causes or experience states are being replaced by expectations about those hidden states and they still cascade down but they do so via the prediction errors that crucially now are returned back up to update and inform the expectations using this gradient flow or Kalman filter or predictive coding equation here so the picture that falls out for free from that gradient flow interpretation of non-equilibrium self-organization - steady state is exactly what we see in the brain this counter stream of ascending prediction errors that induce belief updating or base and belief updating to provide better explanations that can be returned by descending predictions - prediction error cells so there are self cancelling through this recurrent message passing that is hierarchically composed or arranged here's a quick example for the visual system just to sort of put action back into into the story so imagine we have some visual retinal information coming in here against the lateral geniculate nucleus it's in receipt of top-down predictions from the visual cortex and they are partially good but not completely so there's a prediction error that gets

passed up as the news where the information that has yet to be explained that excites changes in the expectations in the visual cortex but crucially these expectations are themselves being predicted by a high level so that's the secondary prediction error and that's passed up and so on so on to increasingly a hierarchical depth until you've minimized prediction error all levels in the hierarchy and you have a set of expectations and different levels of coarse graining in space and in time that provides an account of your sensorial of multiple levels of abstraction I've also put in here there's another sort of sensory information from the sensory states of the Markov bank at the proprioceptive input the status of my muscles my oculomotor system now these could come in they could be in receipt of top-down predictions and I could change my mind about where I am looking but here's where the reflex comes in this is where action comes in there's another way in which these particular prediction errors can be minimized and that's if they couple straight back to the outside world through the active States so all that means is if I register there's a difference between my sensed stretch of my muscle and my predictive stretch of the muscle then I will contract or relax the muscle until the sensations match the predictions so what I've just described as a classic thank you a classical reflex arc and we can use that to simulate all sorts of things I'll just give you one example really to illustrate the difference between this example and the interesting thing which justifies the title of this talk in couple of minutes we've just used that hill-climbing equation with a particular generative model to emulate or simulate slow person tracking I won't go into the matter FATA carcasses of on substitution area don't need to know how this simple Newtonian mechanics are described an interesting model where I believe that the same things that pull a target around are also pulling my eyeball so we've written into this particular simulated subject or synthetic agent the prior belief in their charity model that causes predictions of both what they see and what they feel from their eye movements the belief that there's some fictive invisible point of attraction out there there is pulling a target around and is also pulling their center of gaze and if the agent believes this is the best explanation for its sensory world then it will fulfill those predictions by tracking and also anticipating the location of the target and which is very very realistic slope assume target movements realistic to the extent that you can simulate occlusion so this is just a simulation here with a sinusoidal target here that goes behind an occluder here and then we're looking at the beliefs of this simulated agent during the periods of occlusion denoted by the gray bars here and in particular what would happen if I changed the precision of the gain or the confidence or afforded their prediction errors would it make the sorts of mistakes that people with schizophrenia make and indeed it does and you can simulate both

the psychophysics and also the the the tracking of the target in the slope soup paradigm to emulate very very precisely and in detail the sorts of slow passing deficits you get from a lesion of this active inference formulation of how we sample our world I don't want to talk about this is what I want to talk about in in the last couple of minutes I want to talk about active inference and epistemic affordance so I'm going to make the final move now which is taking us to a more abstract application of these ideas where I'm going beyond just saying I have beliefs about the world and I'm going to act upon the world to make those beliefs come true by vye minimizing my prediction error I'm going to say that my journey immoral now entertains the consequences of acting in the future and furthermore I believe that I am the sort of creature that will act as if I have I am minimizing the free energy that will ensue or the prediction errors that will ensue if I did this all that and I'm now going to select from one policy or another policies purely in the service of minimizing this free energy bound on surprisal if I were talking to philosophers what we're saying is basically if I act I must have beliefs about the consequences of action so if I have posterior beliefs about action I must have pry beliefs about action and therefore it follows that what are those Prai believe well if I am a self evidencing creature in other words if I try to maximize my model evidence therefore I believe I behave as if I am a self evidencing creature which technically means that I will select those actions that minimize expected surprise or self information I free energy consequent upon an action so very briefly what happens is that you set up a bunch of policies actions like where to look next and then you work out technically using the posterior predictive density over the outcomes if I did that the free energy that you would have got if you write that down it actually transpires that it has very intimate relationships with a lot of extant approaches to planning and ultimate control and an economics so the expected free energy has these two bits to it it's expectation over posted predictive density of the sensory outcomes here's the normal here's the evidence bit and here's a cow divergence so I'm going to demonstrate that basically in the context of just choosing where to look next that's that's basically my space of policies that I have to select from so what I'm going to be doing is selecting the place to look next that max minimizes my expected free energy what does that mean it's basically minimizing expected surprise which means and minimizing uncertainty which means maximizing information gain reducing as much uncertainty as I can by selecting those epistemic lis rich sensory or what I believe will be an aesthetically rich parts of the world that will reduce as much uncertainty as possible about about what I think is causing it and indeed if you compute this sort of expected free energy create salience map so we've actually referred to this expected

free energy or an important component of it as salience it's also known as Bayesian surprise this was just making the point that we can actually go into this expected free energy expression and just focus in on various mixtures of its terms and pull out stuff which has been in literature for decades so if I ignore the pragmatic or the prior bleeds my pry preferences I have no preference about what I'm going to see then what is left is earth what atium Baldy will call basin surprise what Horace baller would call mutual information so if I take it preferences out of the game then I end up with basically an info max or minimum redundancy principle but it still is only part of the story I can put the Preferences back in but I can retain uncertainty about the outcomes of action and then I get risk sensitive control of people optimal control theory use where they minimize the divergence between preferred sensations or outcomes and those that I predict under a particular outcome so if you're an economist this would also be resuscitated choice behavior in economics and I can take the final sort of uncertainty out and I can actually just get back to expected utility theory or enforcement learning so I take all sorts uncertainty off the table both in terms of what would happen if I did that and my uncertainty about the state of the world we get back to good old-fashioned reinforcement learning and expected utility theory and this is the final slide that just shows you the solutions to this gradient flow with a particular Journal to model where it's just trying to resolve uncertainty about the causes of these sensations these four build samplings that you're seeing here under the hypothesis that could be an upright phase a side of a face or an inverted face and how it's choosing by appeal to the 6:53 energy or salience map the best bits of the image to look at to resolve as much uncertainty as it can so there were a few moves it's pretty certainly wasn't a bright face and indeed it was an upright face so I will conclude by giving the last word as always to Helmholtz who basically said what I've just said that said in one sentence each movement we make by which we alter the the appearance of objects should be thought of as an experiment designed to test whether we have understood correctly the invent relations of the phenomena before us that is their existence in definite spatial relations and with that I just want to thank all the people whose ideas I've been talking about and of course thank you for your attention thank you very much indeed [Applause] questions are you well it's a nice gentle question not very excitement is the word worry is more and it's usually worried about their gender the people I supervise so at the moment the it seems to be taken towards natural language processing and natural language understanding that the people I work with a cognitive neuroscientist centered in speech and noise and and hearing and recovery of language function following stroke on the other side of the coin there are people

who want to see if these ideas scale up in a commercial setting so they're going off trying to you know get funding for startups and the likes of that that agender is really finding toy proofs of principle where you get these sorts of schemes outperforming deep learning and open G open AI gym formats and the like so that's that's not exciting but it's worrying I think so I should also say a large number of my younger colleagues or also come from ecology and ethology and theoretical biology so they've taken these ideas and made them work in the context of eco nation construction and mathematical modeling of sort of not selection but the sorts of processes you'd see in new wave evolutionary psychology and deal where the cultures come in and how does this bootstrapping to in the inside and the outside States where the outside states are other creatures like me get into the game of making the world much more predictable and learn and also that that's also very interesting and I've mentioned already some people might 11 at Tufts is taking these ideas and seeing if we can use them to explain games and dances and self-organization active inference at the level of cells and whether that is an explanation for pattern formation in terms of morphogenesis and possibly in the context of near plays or in cancer so there are lots of exciting applications very P bases that were just worries as a basis of excitement with whether we can keep up or but I think we will order more yes yes oh well it lives center stage you know so this notion of epistemic affordance that you get from these simulations where this is just about leave appalling this is basically what would happen if I did if I did that this I think is one of the most beautiful expressions of this imperative to reduce uncertainty so remember mathematically free energy upper bound surprised and expected surprise is uncertainty is entropy so all that we are saying here is that we are things that act to minimize uncertainty now one of the best ways to act to minimize uncertainties to see what happened if I did that so when you look at the posterior beliefs over the consequences of action what would happen if I pull that lever that becomes incredibly attractive from the point you're in the jargon of Gibsonia people the the in activist movement it has epistemic affordance so this Bayesian surprised or this this mutual information if you're Horace Barlow I think that's probably more easily understood as this this intense epistemic affordance or attractiveness so the yellow lever also they believe in the Yellow Submarine if it has not been pulled before all other things being equal is the most has the greatest affordance to because it's novel so when you rewrite this base and surprise not in terms of posterior beliefs about states of the world but personally I believe it's about continues that the parameters that generate states in the world the consequences of action that looks exactly like novelty so that question I think is a great question on two levels first of all it allows me just to

sort of D technical eyes the basic imperative and we're talking about it's just about sampling salient stuff be just novelty seeking but on another level you're absolutely right as well that you know these imperatives do describe people and particular scientists you know this is just the you know describing exactly what we do always spend our entire lives doing is generating hypotheses carefully designing these experiments with epistemic affordance to disclose and reveal the most evidence that we can and if we could scientists we make these good every stomach moves and if we're not we don't get funding and we confirm our own ideas or get ignored so really this is that this is another way of disarticulated the the notion of the the brain as a scientist you know the both subject exactly the same the same the same notions and that's a tricky question to answer in terms of normally I would have a simulation to show you how it works reproduction here is is one way of if you like a particular class of having these attendant attracting sets so one level from the point of view of a species over a certain period of time before after the speciation occurred to create that species you can write down the trajectory of the species where each of those cycles I was protecting before as my daily life cycle or my heartbeat that could actually be over better than an evolutionary time scale so you can look at that as you know that as revisiting certain states of being over somatic time like being an embryo being and as I go being an infant and go through a neurodevelopmental cycle so the key question there is do the same mechanics apply at the level of evolution that must be in play if there is an attracting set that is a species and that leads us into the game of now reinterpreting the price equation or the replicator dynamics that are used to model evolution not so much specifically it as a reproduction but certainly in terms of you know evolutionary theory more generally as basically a process of inference and that's very interesting because there are a lot of there's a lot of mathematically the answer is yes absolutely and I think that chuckle yes Frank has written about that and John Campbell talking about Universal Darwinism it's a truism that the population dynamics that are used to model theoretical evolution are just Kalman filters they are just variational free energy minimizing great gradient flows the more interesting thing I think is the the memes and the rhetoric you can put on top of that so now if that is true it means a natural selection now becomes Bayesian model selection and now beliefs become or internal states now become phenotypes so the phenotype now becomes if you like a statement of belief about what sorts of things are fit for purpose in this eco nation and the Bayesian model alert selection now does a structure learning of the phenotype where you've actually got explicitly an ensemble density encoding the posterior belief so evolution now becomes a scientist so evolution now does its belief updating using a

sample density over the phenotypes that constitute the population density that conforms to exactly the same rules that the Fokker-Planck equation describes so I think it's a lovely game to be played in terms of theoretical biology just unpacking the same maths that you might use to describe you know it's creating a wave equation but applying it to population dynamics and just see what you mean when you talk about natural selection and how that means relates to things like structure learning it's a natural language processing for example example mathematically they all have the same form that was a bit of a hand wavy answer at the level of reproductions the reproductions an interesting one because it implies a splitting of the genotype and that's that I think requires some finer or more detailed analysis